

Technology Stack

Date	30 June 2026
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build the solution. A well-chosen technology stack ensures the application is scalable, maintainable, and aligned with project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	OptiCrop's interface (soil/climate input forms, the Crop Feasibility Analyzer, and results dashboard) is built with lightweight HTML/CSS/JS for a fast, mobile-responsive, easy-to-navigate UI that farmers can use directly from a phone, without the overhead of a heavier frontend framework.
2	Backend / Server-Side	Python with Flask	Flask provides a lightweight, easy-to-integrate web framework that connects the user-facing forms directly to the Python-based Machine Learning pipeline (NumPy, Pandas, Scikit-learn), enabling quick development and straightforward deployment of the prediction API.
3	Database / Data Storage	CSV-based dataset / SQLite (with option to scale to PostgreSQL)	The crop recommendation models are trained on structured agricultural datasets (N, P, K, temperature, humidity, pH, rainfall, crop label); SQLite is used for lightweight storage of user submissions and results, with the option to scale to PostgreSQL as data volume grows.
4	Cloud / Hosting / Deployment	Render	Render offers simple, cost-effective hosting for the Flask web application (currently live at opticrop.onrender.com), with easy continuous deployment from the source repository and minimal DevOps overhead for a student/research project.

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5	Version Control & CI/CD	Git & GitHub	GitHub enables seamless collaboration between team members on the codebase and datasets, version history tracking, and integrates directly with Render for automatic redeployment whenever changes are pushed.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib/Seaborn (ML & data tooling)	Scikit-learn provides the Logistic Regression, KNN, Decision Tree, Random Forest, and K-Means algorithms that power the Multi-Algorithm ML engine; Pandas/NumPy handle data preprocessing, and Matplotlib/Seaborn support visual analytics such as the Climate Adaptive Mapping and Soil Nutrition Profiling insights.